Measurement and Instrumentation for Energy Systems

Module 2: Sensor Classification, Static and Dynamic Characteristics, Sensors for Energy Measurements such as Strain, Pressure, Level, Flow, Temperature, Density, Humidity, Viscosity, PH, Emissions, etc.

Introduction to Sensors

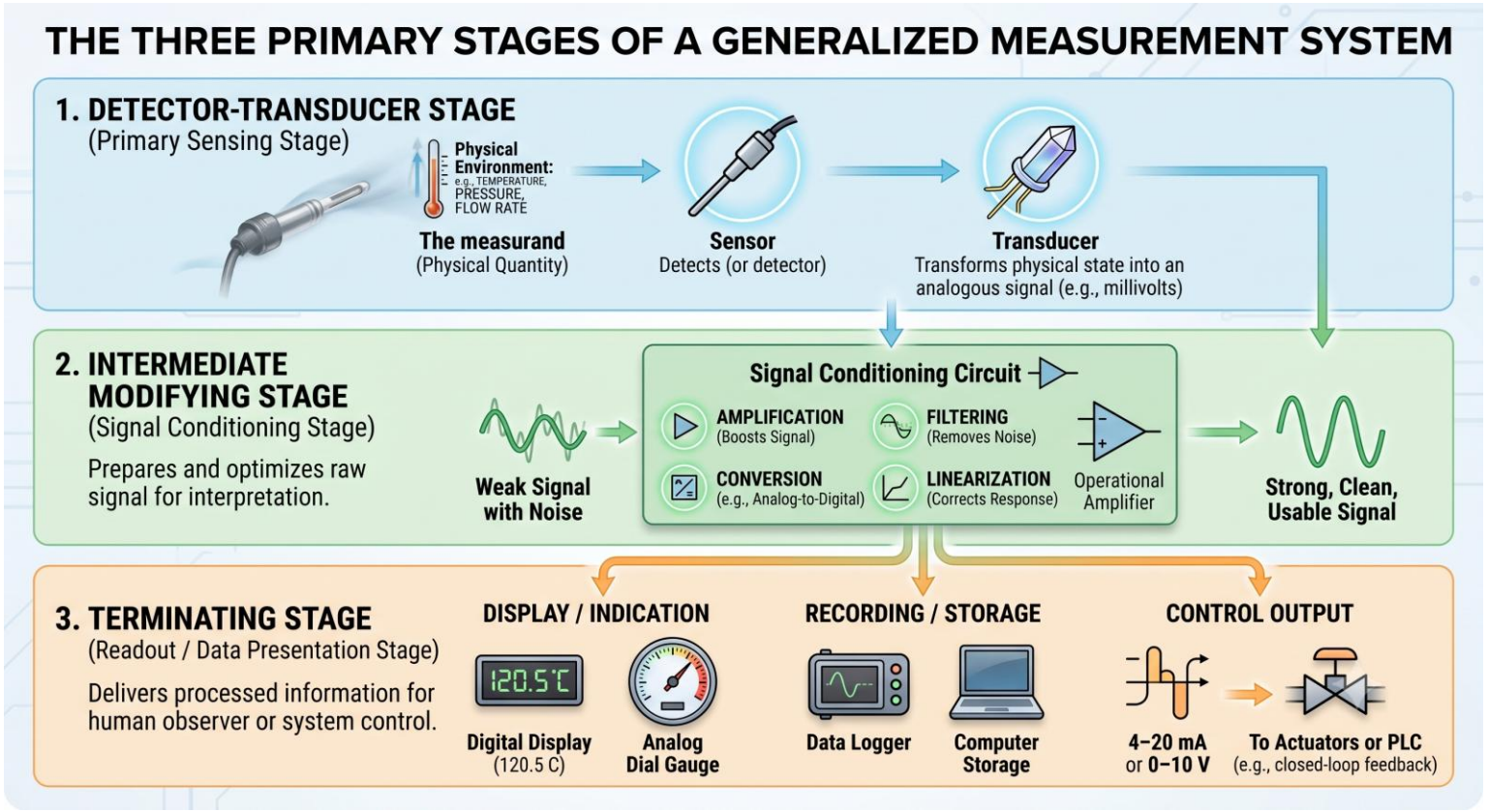
Sensor (or detector): The primary component in a measurement system that directly detects a physical variable (such as temperature, pressure, or motion).

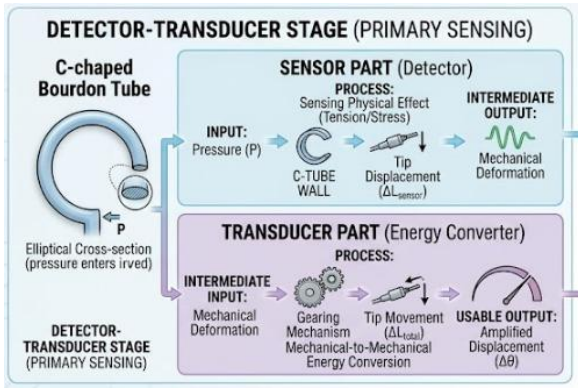
Transducer: A device which converts the detected physical variable into a more usable mechanical or electrical signal.

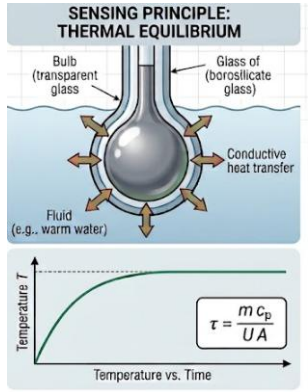
Examples:

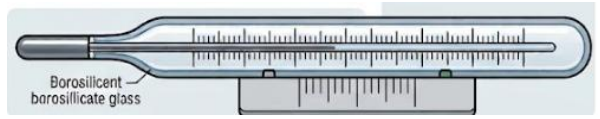
Thermometer: Acts as a temperature-sensing device that detects thermal states and converts them into an indicated temperature reading.

Bourdon-Tube Pressure Gage: The oval cross-section Bourdon tube acts as a detector-transducer stage that senses an input pressure source and transforms it into a mechanical displacement of the tube.









Introduction to Sensors

Difference Between Sensors and Transducers

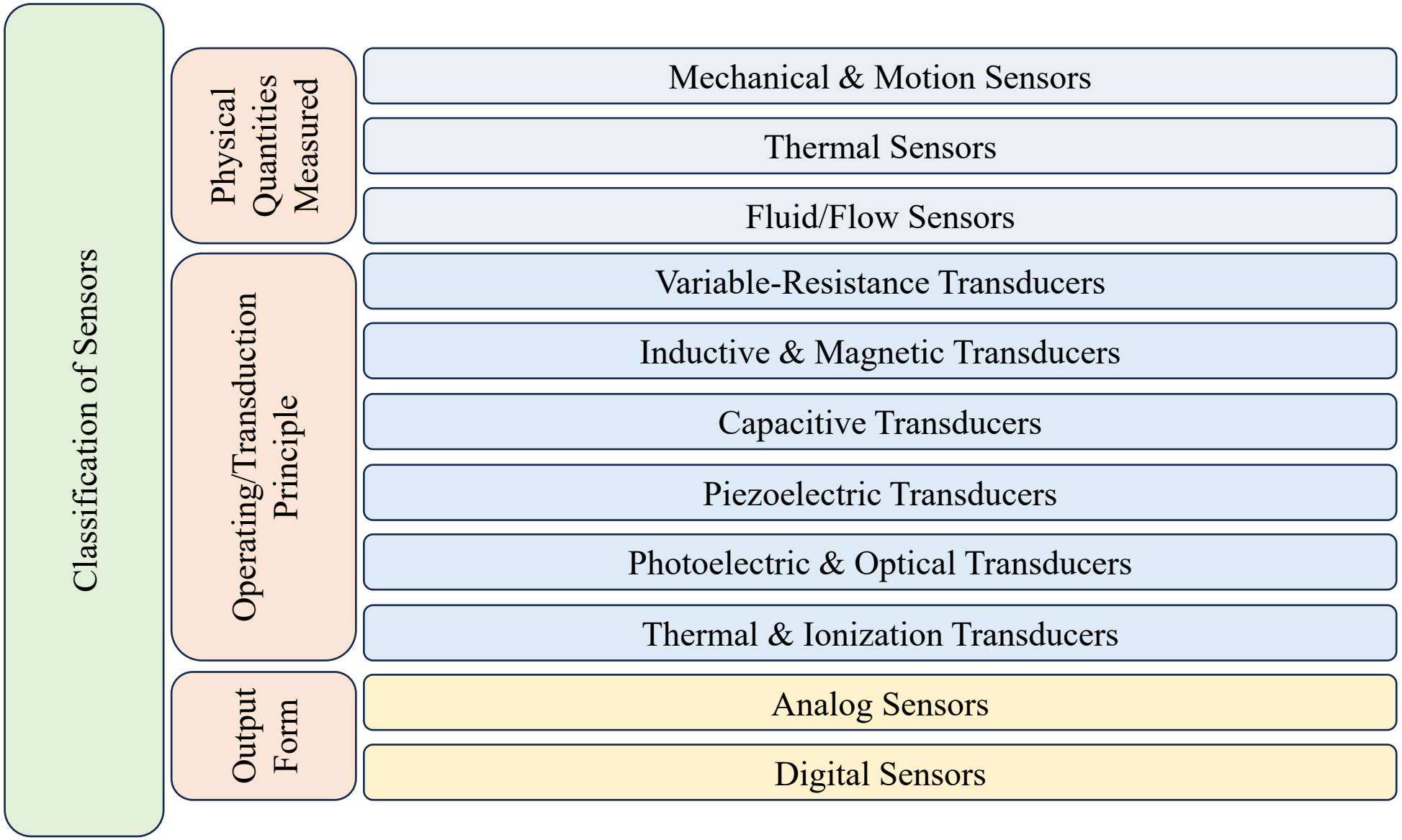
Sensor	Transducer
The sensing element that directly detects and responds to a physical quantity or variable from the environment.	A transducer is any physical device or element that converts a signal from one form into another. In instrumentation, measurement systems, and control engineering, it specifically converts a non-electrical or physical quantity (the measurand, such as pressure, temperature, light, or force) into a usable output signal - most commonly an electrical signal or a mechanical displacement.

Examples:

Bourdon-Tube Pressure Gage: The oval cross-section Bourdon tube acts as a detector-transducer stage that senses an input pressure source and transforms it into a mechanical displacement of the tube.

Thermometer: Acts as a temperature-sensing device that detects thermal states and converts them into an indicated temperature reading.

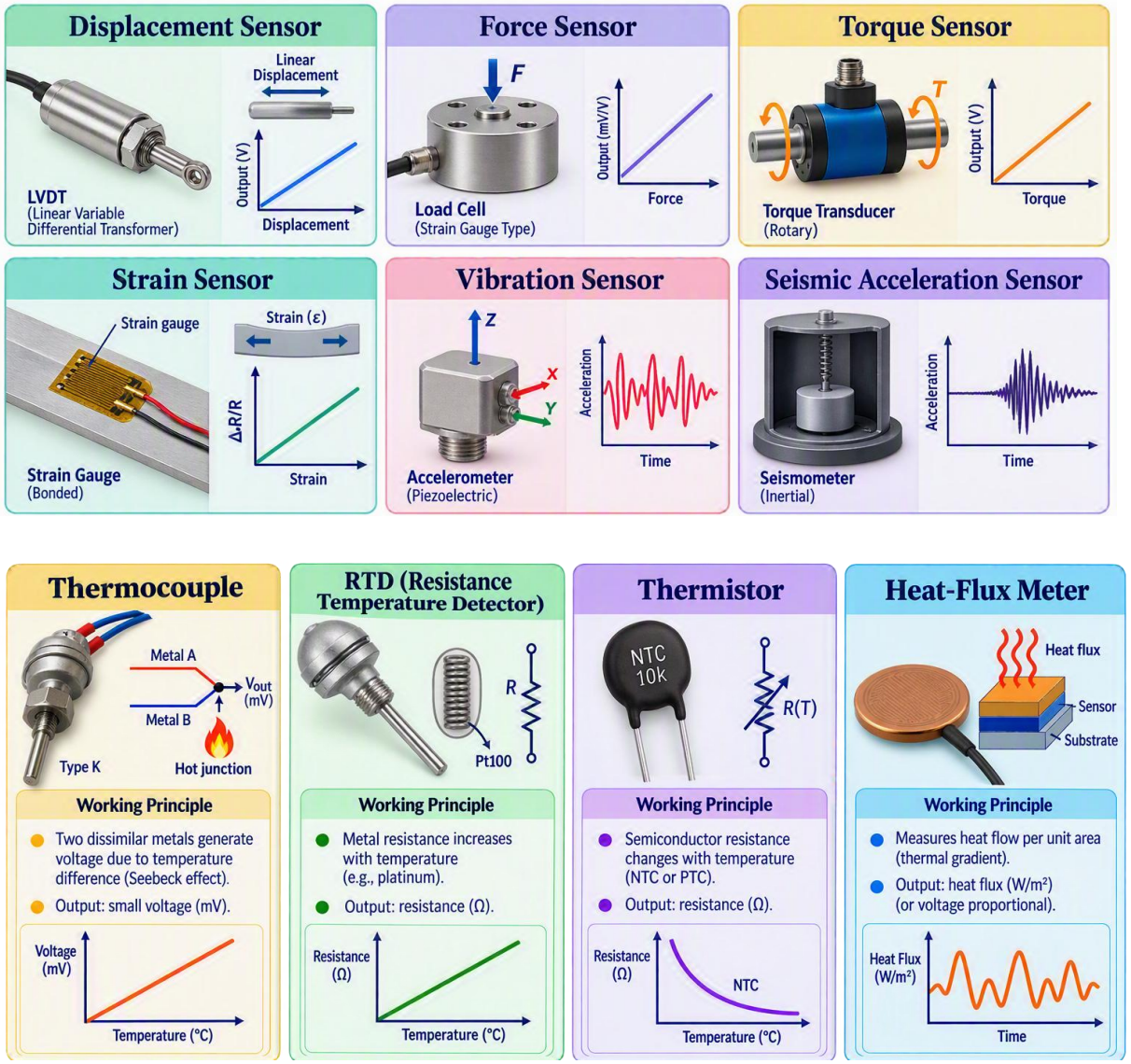
Classification of Sensors/Transducers



Sensor/Transducer Classification

Classification by Physical Quantities Measured (Application Domain):

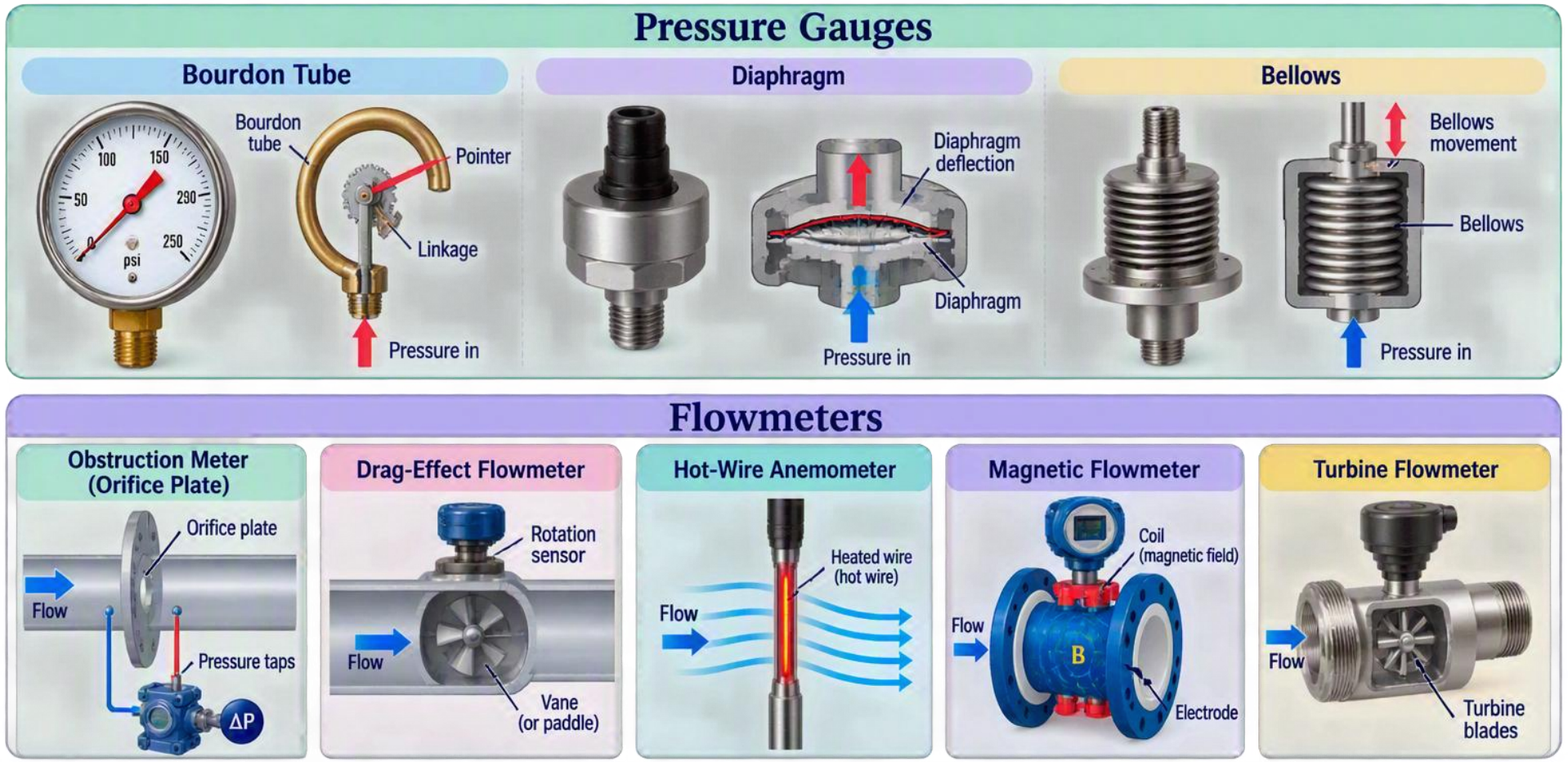
- Mechanical & Motion Sensors: Displacement, force, torque, strain, vibration, and seismic acceleration.
- Thermal Sensors: Temperature measuring elements (thermocouples, resistance temperature detectors/RTDs, thermistors) and heat-flux meters.
- Fluid/Flow Sensors: Pressure gauges (Bourdon tubes, diaphragms, bellows) and flowmeters (obstruction meters, drag-effects, hot-wire anemometers, magnetic flowmeters).



Sensor/Transducer Classification

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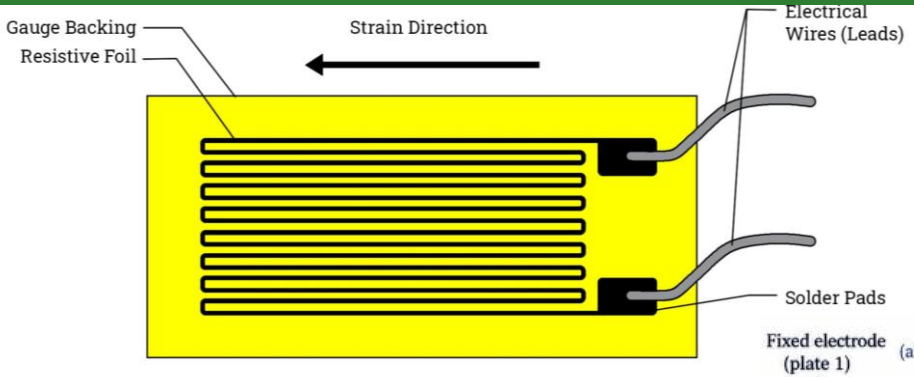
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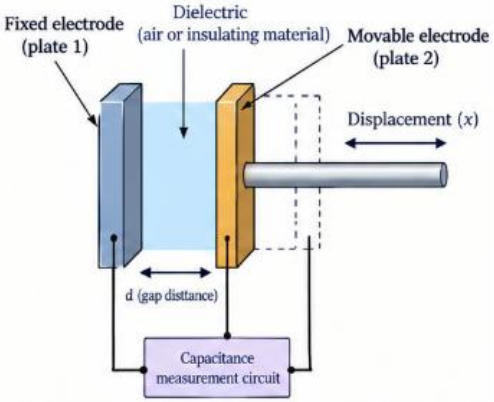
Sensor/Transducer Classification

Classification by Operating/Transduction Principle (Electrical & Physical Transducers):

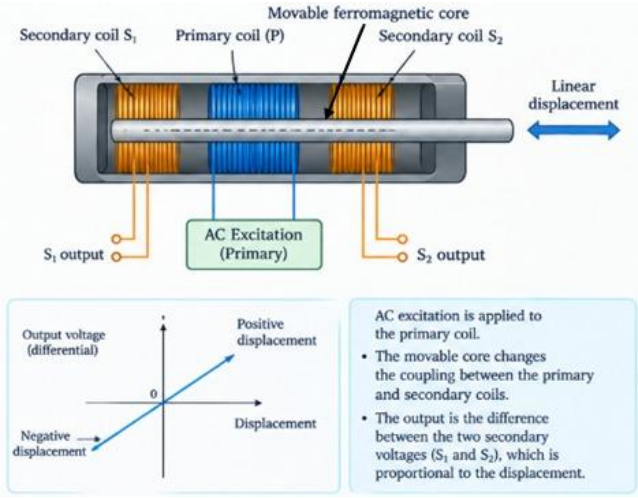
- Variable-Resistance Transducers: Devices where a physical parameter (such as displacement or strain) alters electrical resistance (e.g., strain gages, potentiometers).
- Inductive & Magnetic Transducers: Devices relying on electromagnetic induction or changes in magnetic flux (e.g., Linear Variable Differential Transformers - LVDTs, Hall-effect transducers, magnetometer search coils).
- Capacitive Transducers: Devices that measure changes in electrical capacitance caused by spacing or area variations between plates.



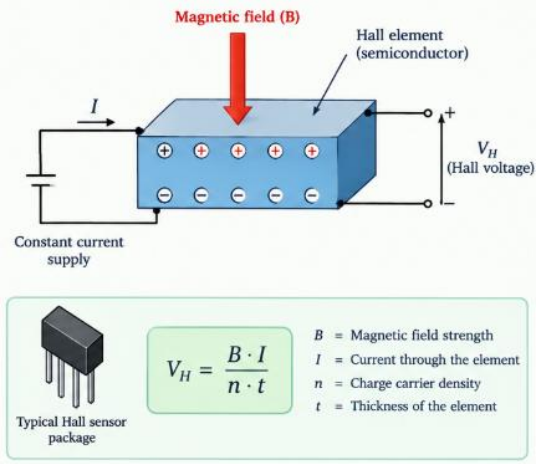
Strain-gauge construction



Capacitive transducer



LVDT

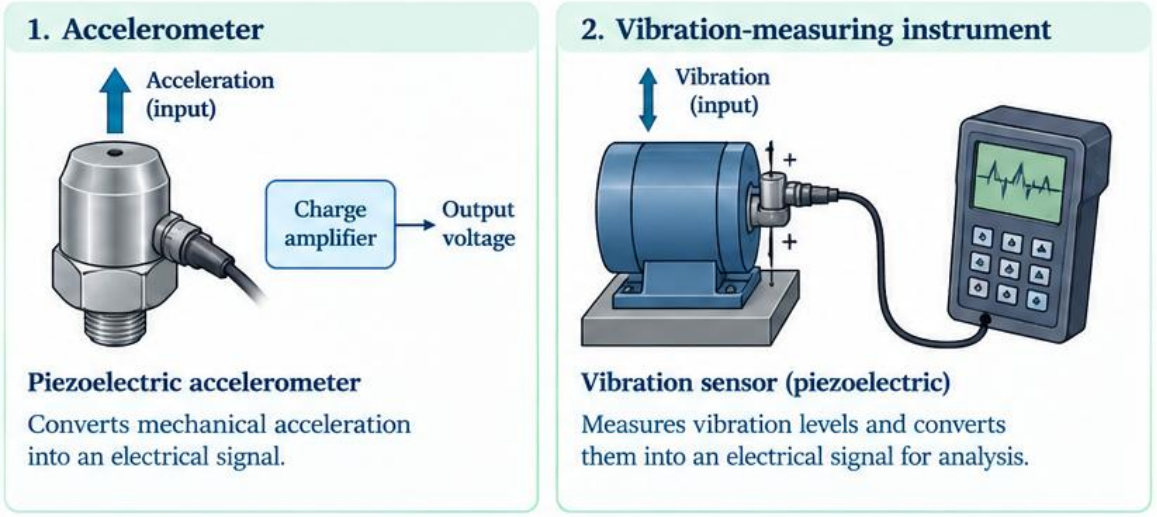


Hall-effect transducer

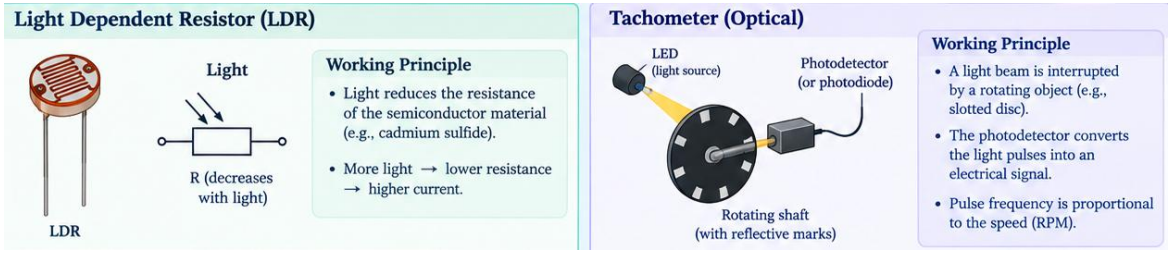
Sensor/Transducer Classification

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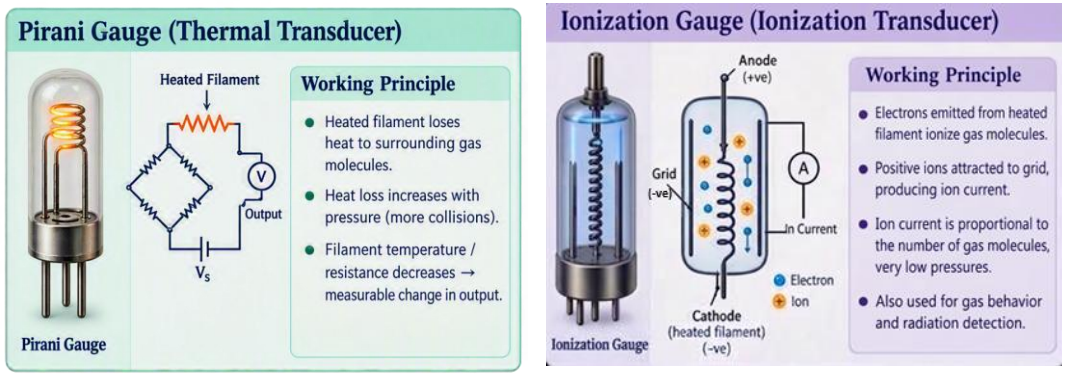
- Piezoelectric Transducers: Active devices that generate an electrical charge or voltage when subjected to mechanical stress, force, or acceleration (Accelerometers and vibration-measuring instruments).
- Photoelectric & Optical Transducers: Devices responding to light parameters, classified further by effects like photoconductive, photovoltaic, and photoemissive behaviors (Light-sensitive resistors (LDRs), Tachometers).
- Thermal & Ionization Transducers: Specialized sensors for measuring low pressures, gas behaviors, or radiation characteristics (e.g., Pirani gauges, ionization gauges).



Piezoelectric transducers



Photoelectric & Optical Transducers

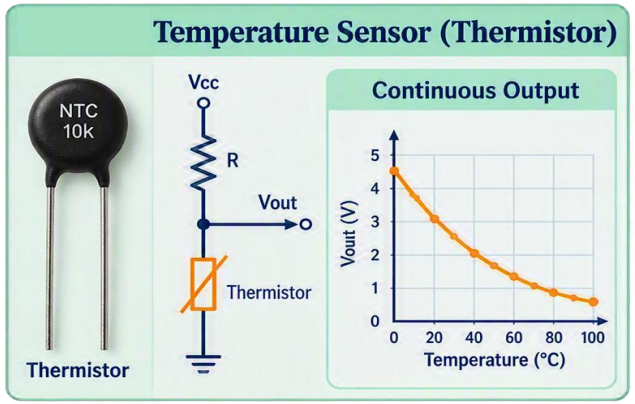


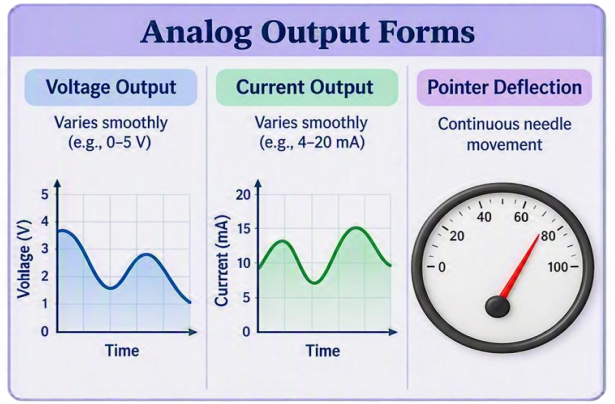
Thermal & Ionization Gauges

Sensor/Transducer Classification

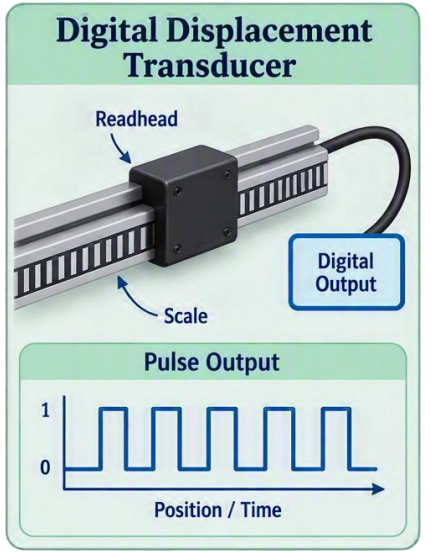
Classification by Output Form:

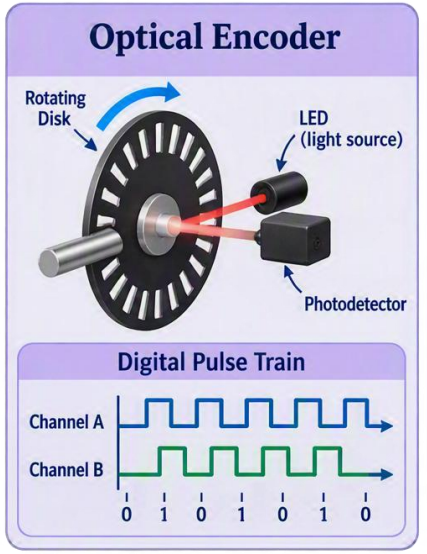
- Analog Sensors: Output a continuously varying physical or electrical signal (such as voltage, current, or pointer deflection).
- Digital Sensors: Output discrete or pulse-coded signals (such as digital displacement transducers, optical encoders).





Analog Sensors





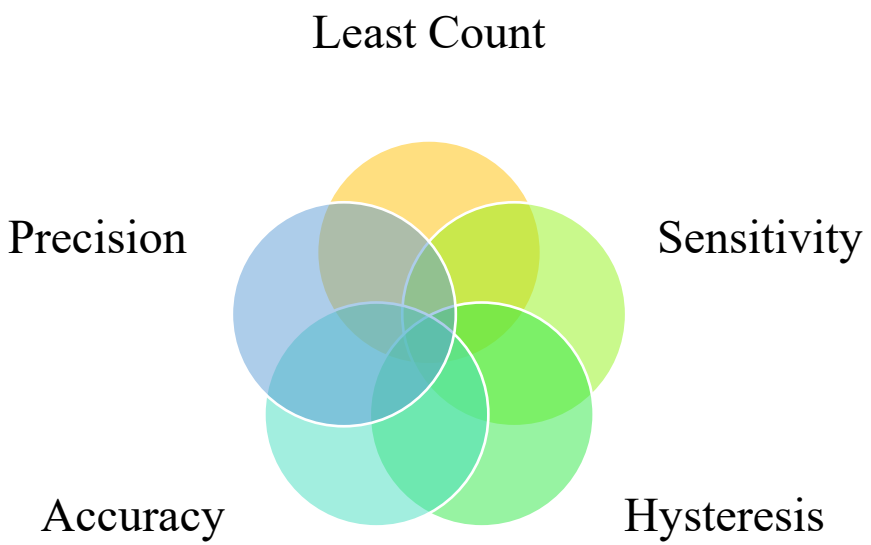
Digital Sensors

Static & Dynamic Characteristics

- Static Characteristics: Static characteristics describe how an instrument behaves when the measured quantity is constant or changes slowly enough that time-response effects are not important.

  In other words, they describe the sensor's performance under approximately steady-state conditions.

The main static characteristics are:



Accuracy: The closeness of agreement between a measured value and the true or accepted reference value.

Precision: The degree of mutual agreement or repeatability among independent measurements made under identical conditions.

Sensitivity: The ratio of the incremental change in output signal to the corresponding incremental change in the physical input quantity being measured.

Least Count: The smallest increment or change in the measured quantity that an instrument can reliably detect and display.

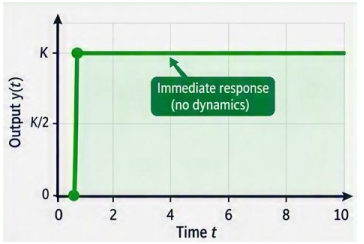
Hysteresis: The maximum difference in output for the same input value when approached from opposite directions (increasing versus decreasing inputs).

Static & Dynamic Characteristics

- Dynamic Characteristics: Dynamic characteristics become important when the measured quantity changes with time.

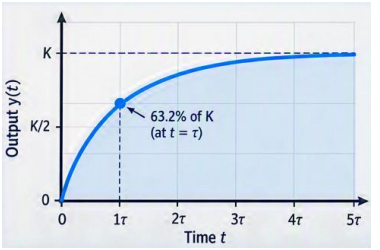
The three specific types of dynamic characteristics are:

- **Zero-Order systems:** The instrument output moves immediately to a new steady value following a step change in the measured quantity. There is essentially no time lag between the input change and the output response, as seen in devices like a motion-measuring potentiometer.



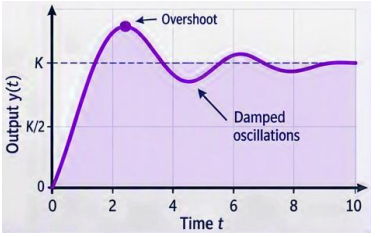
$$a_0\, y(t) = b_0\, x(t)$$

- **First-Order systems:** A first-order system is a measurement system whose dynamic behavior is described by a first-order differential equation. This behavior is defined by a time constant (t), which is the time required for the instrument's output to reach 63.2% of its final value. Standard thermocouples and thermometers exhibit a first-order response, where the temperature reading rises exponentially rather than instantaneously when plunged into a hot environment.



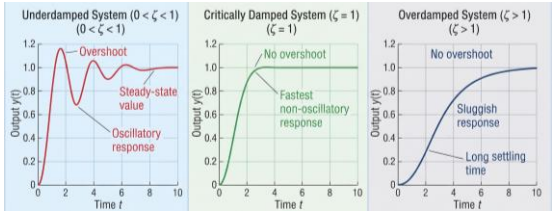
$$a_1 \frac{dy(t)}{dt} + a_0\, y(t) = b_0\, u(t)$$

- **Second-Order systems:** A second-order system is a measurement system whose dynamic behavior is described by a second-order differential equation. Second-order systems are associated with effects such as mass inertia and damping. A common mechanical example is a spring–mass–damper system, which can represent pressure or force transducers.



$$a_2 \frac{d^2y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0\, y(t) = b_0\, x(t)$$

Static & Dynamic Characteristics

Feature	Zero-Order Systems	First-Order Systems	Second-Order Systems
Governing Differential Equation	$a_0\, y(t) = b_0\, x(t)$	$a_1\, (dy(t)/dt) + a_0\, y(t) = b_0\, x(t)$	$a_2\, (d^2y(t)/dt^2) + a_1\, (dy(t)/dt) + a_0\, y(t) = b_0\, x(t)$
Standard Canonical Form	$y(t) = K \cdot x(t)$	$\tau\, (dy(t)/dt) + y(t) = K \cdot x(t)$	$(1/\omega_n^2)\, (d^2y(t)/dt^2) + (2\zeta/\omega_n)\, (dy(t)/dt) + y(t) = K \cdot x(t)$
Key Parameters	• Static sensitivity (K)	• Static sensitivity (K) • Time constant (τ)	• Static sensitivity (K) • Undamped natural frequency (ω_n) • Damping ratio (ζ): Damping present in the system/ Critical damping
Energy Storage Elements	None (pure resistive dissipation or instantaneous ratio scaling).	Single energy storage element (e.g., thermal mass, single electrical capacitance).	Two interacting energy storage elements exchanging kinetic and potential energy (e.g., mass-spring, inductor-capacitor).
Step Response Function y(t) **[for x(t) = U(t)]**	$y(t) = K$	$y(t) = K \cdot (1 - e^{(-t/\tau)})$ 	Depends on damping ratio (ζ): • $\zeta < 1$ (Underdamped): Oscillatory decay toward steady-state. • $\zeta = 1$ (Critically Damped): Fastest non-oscillatory rise. • $\zeta > 1$ (Overdamped): Sluggish sum of two decaying exponentials.
Dynamic Response Characteristics	Instantaneous response; zero dynamic error; no amplitude attenuation or phase lag.	Asymptotic exponential rise: • 63.2% at $t = \tau$, • 86.5% at $t = 2\tau$, • 95.0% at $t = 3\tau$, • 99.3% at $t = 5\tau$, No overshoot.	Response dynamic profile governed by ζ: • Underdamped: exhibits overshoot, ringing, and peak time. • Dynamic error is frequency-dependent with resonance near ω_n.
Typical Transducer Examples	Linear / rotary potentiometers, Strain gauge Wheatstone bridges	Mercury-in-glass thermometers, Bare thermocouples, Thermistors, RC low-pass filters	Piezoelectric accelerometers, Elastic diaphragm pressure sensors, Seismic pickups, D'Arsonval galvanometers

Strain Measurement

Strain is a measure of the geometric deformation experienced by a material under mechanical loading, defined as the ratio of the change in length (ΔL) to the original, undeformed length (L) :

$$\epsilon = \frac{\Delta L}{L} = \frac{dL}{L}$$

Poisson's Ratio (μ) : Defined as the ratio of transverse unit strain to axial unit strain

$$\mu = -\frac{\epsilon_t}{\epsilon_a} = -\frac{dD/D}{dL/L}$$

Typical Elastic Value : $\mu \approx 0.3$ for many common materials.

Plastic State (Constant Volume, dV = 0):

Volume conservation yields $\frac{dA}{A} = -\frac{dL}{L}$

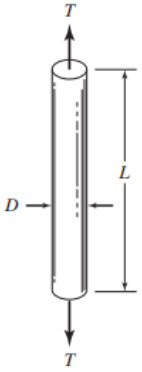
which translates in terms of diameter to $2\frac{dD}{D} = -\frac{dL}{L}$.

Results in $\mu = 0.5$ under plastic conditions.

Young's Modulus formula (derived from Hooke's Law):

Hooke's law of elasticity states that within the elastic limit of a material, the stress applied to the material is directly proportional to the resulting strain:

$$E = \frac{\sigma}{\epsilon} = \frac{F/A}{\Delta L/L}$$



where:

E is Young's modulus (units of pressure or stress, such as Pascals, F is the applied normal force, A is the initial cross-sectional area perpendicular to the force.

Strain Measuring Gages:

Electrical Resistance Strain Gauges	Mechanical Extensometers
Optical Fiber Sensors	Digital Image Correlation (DIC)
Capacitive Strain Gauges	Photoelasticity
Piezoelectric Strain Sensors	Ultrasonic and Acoustic Methods

Electrical-Resistance Strain Gages

Principle: Electrical resistance of a conductor changes when it is subjected to mechanical deformation.

- An electric conductor is bonded to the specimen with an insulating cement under no-load conditions.
- A load is then applied, which produces a deformation in both the specimen and the resistance element.
- This deformation is indicated through a measurement of the change in resistance of the element.
- The resistance of the conductor is

$$R = \frac{\rho L}{A} \tag{1}$$

Where ρ = Resistivity of conductor (Ω-m)

Differentiating,

$$dR = \frac{L}{A}d\rho + \frac{\rho}{A}dL - \frac{\rho L}{A^2}dA \tag{2}$$

Dividing by $R = \frac{\rho L}{A}$ on both the sides,

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dL}{L} - \frac{dA}{A} \tag{3}$$

Since, Area $A = \frac{\pi D^2}{4}$ (4)

Differentiating,

$$dA = \frac{\pi}{4} \times 2DdD \tag{5}$$

Dividing by $A = \frac{\pi D^2}{4}$ on both the sides,

$$\frac{dA}{A} = 2\frac{dD}{D} \tag{6}$$

Substituting value of $\frac{dA}{A}$ in equation (3)

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dL}{L} - 2\frac{dD}{D} \tag{7}$$

Axial strain

$$\epsilon_a = \frac{dL}{L} \tag{8}$$

Transverse strain

$$\epsilon_t = \frac{dD}{D} \tag{9}$$

Putting the values of $\frac{dL}{L}$ and $\frac{dD}{D}$ in equation (7)

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \epsilon_a - 2\epsilon_t \tag{10}$$

Electrical-Resistance Strain Gages

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \epsilon_a\left(1 + 2\left(-\frac{\epsilon_t}{\epsilon_a}\right)\right) \qquad (11)$$

$$\text{Where, } \mu = -\frac{\epsilon_t}{\epsilon_a} \qquad (12)$$

Equation (11) becomes,

$$\frac{\mathrm{dR}}{\mathrm{R}} = \frac{\mathrm{d}\rho}{\rho} + \epsilon_a(1 + 2\mu) \qquad (13)$$

The gage factor F is defined by

$$\mathrm{F} = \frac{\mathrm{dR/R}}{\epsilon_a} = 1 + 2\mu + \frac{1}{\epsilon_a}\frac{\mathrm{d}\rho}{\rho} \qquad (14)$$

Hence, the local strain in terms of the gage factor, the resistance of the gage, and the change in resistance with the strain:

$$\epsilon = \frac{1}{\mathrm{F}}\frac{\Delta \mathrm{R}}{\mathrm{R}} \qquad (15)$$

Material resistivity changes can cause F to range widely from -140 to +175.

If the resistivity of the material does not vary with the strain.

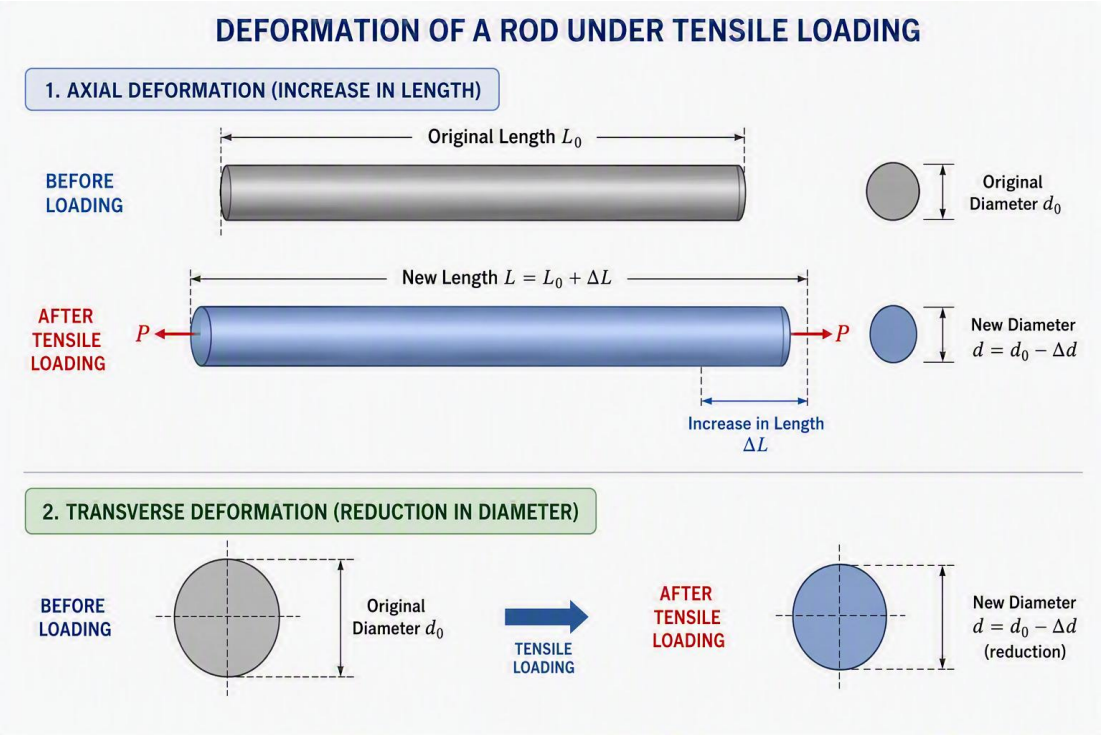
$$\mathrm{F} = 1 + 2\mu \qquad (16)$$

Gage Factor (F) Fundamentals:

- For $\mu = 0.3$, F ≈ 1.6 when resistance changes depend solely on physical dimensions.
- In the plastic region ($\mu = 0.5$, constant resistivity), F approaches 2.

Practical Importance:

- Higher gage factors produce larger resistance changes (ΔR) for a given strain, reducing the need for highly sensitive readout circuitry.

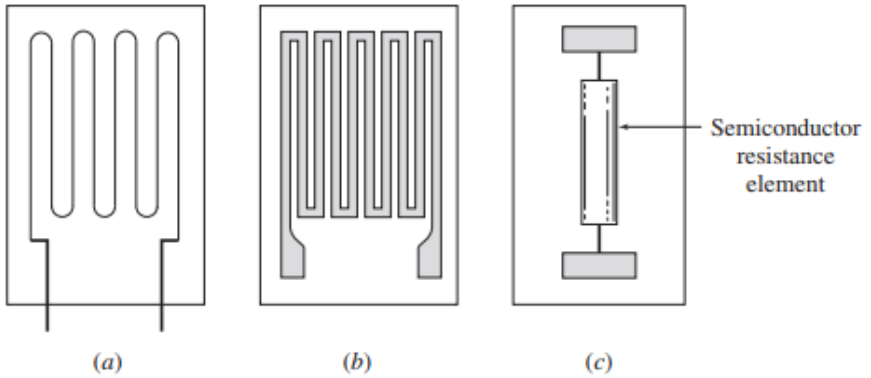


Strain Measurement

Base Length: The finite length over which average strain is measured.

Deformation sensitivity: Minimum deformation that can be indicated by the appropriate gage.

Strain sensitivity: Minimum deformation that can be indicated by the gage per unit base length.

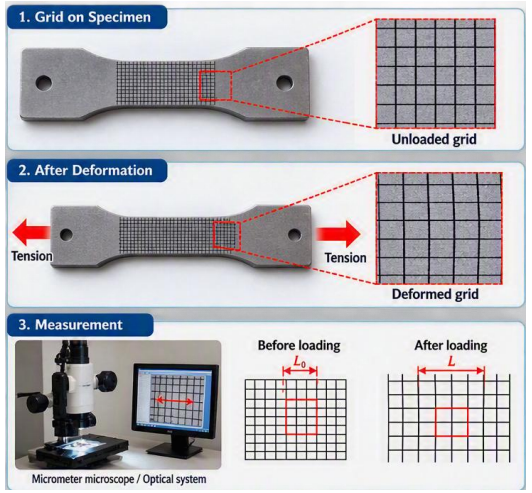


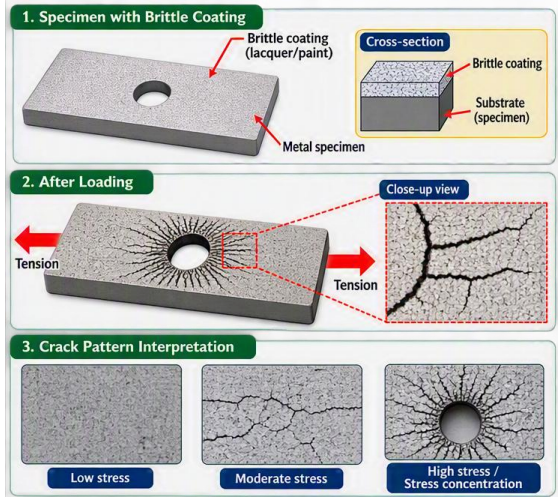
Three types of resistance strain gages. (a) Wire gage; (b) foil gage; (c) semiconductor gage.

Measurement Techniques:

Grid Methods:

- Involves scribed, inked, or photoetched grids on the workpiece measured via micrometer microscopes or photography.
- Best suited for materials and processes with large/appreciable deformations (e.g., sheet-metal forming or rubber models).





Brittle Coatings:

- The specimen is coated with a brittle substance that cracks when specific tensile stress thresholds are reached.
- Ideal for visualizing overall stress distributions and evaluating stress concentrations where standard strain gages cannot easily be installed.

Strain Measurement (Numerical)

A metallic resistance strain gage is bonded to a steel rod having:

Original length, $L = 100$ mm

Original diameter, $D = 10$ mm

Original resistance, $R = 120\ \Omega$

Poisson's ratio, $\mu = 0.30$

Resistivity remains constant during loading ($d\rho = 0$)

When the rod is subjected to a tensile load:

Length increases by $\Delta L = 0.08$ mm

Determine:

Axial strain (ε_a)

Transverse strain (ε_t)

Change in diameter (ΔD)

Gage Factor (F)

Change in resistance (ΔR)

Final resistance of the strain gage

Calculate Axial Strain:

$$\varepsilon_a = \frac{\Delta L}{L}$$

$$\varepsilon_a = \frac{0.08}{100} = 0.0008$$

Calculate Transverse Strain:

$$\mu = -\frac{\varepsilon_t}{\varepsilon_a}$$

$$\varepsilon_t = -\mu\varepsilon_a = -(0.30)(0.0008)$$

$$\varepsilon_t = -0.00024$$

Calculate Change in Diameter:

$$\varepsilon_t = \frac{\Delta D}{D}$$

$$\Delta D = \varepsilon_t D$$

$$\Delta D = (-0.00024)(10)$$

$$\Delta D = -0.0024 \text{ mm}$$

The new diameter is

$$D_f = 10 - 0.0024$$

$$D_f = 9.9976 \text{ mm}$$

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Transverse strain (ε_t)

Change in diameter (ΔD)

Gage Factor (F)

Change in resistance (ΔR)

Final resistance of the strain gage

Calculate Gage Factor:

$$F = 1 + 2\mu$$

$$F = 1 + 2(0.30)$$

$$F = 1.6$$

Calculate Change in Resistance:

$$\varepsilon_a = \frac{1}{F}\frac{\Delta R}{R}$$

Rearranging,

$$\Delta R = FR\varepsilon_a$$

$$\Delta R = (1.6)(120)(0.0008)$$

$$\Delta R = 0.1536\ \Omega$$

Calculate Final Resistance:

$$R_f = R + \Delta R$$

$$R_f = 120 + 0.1536$$

$$R_f = 120.1536\ \Omega$$

Pressure Measurement

Pressure is defined as the force per unit area exerted by a fluid on a containing wall:

$$P = \frac{F}{A}$$

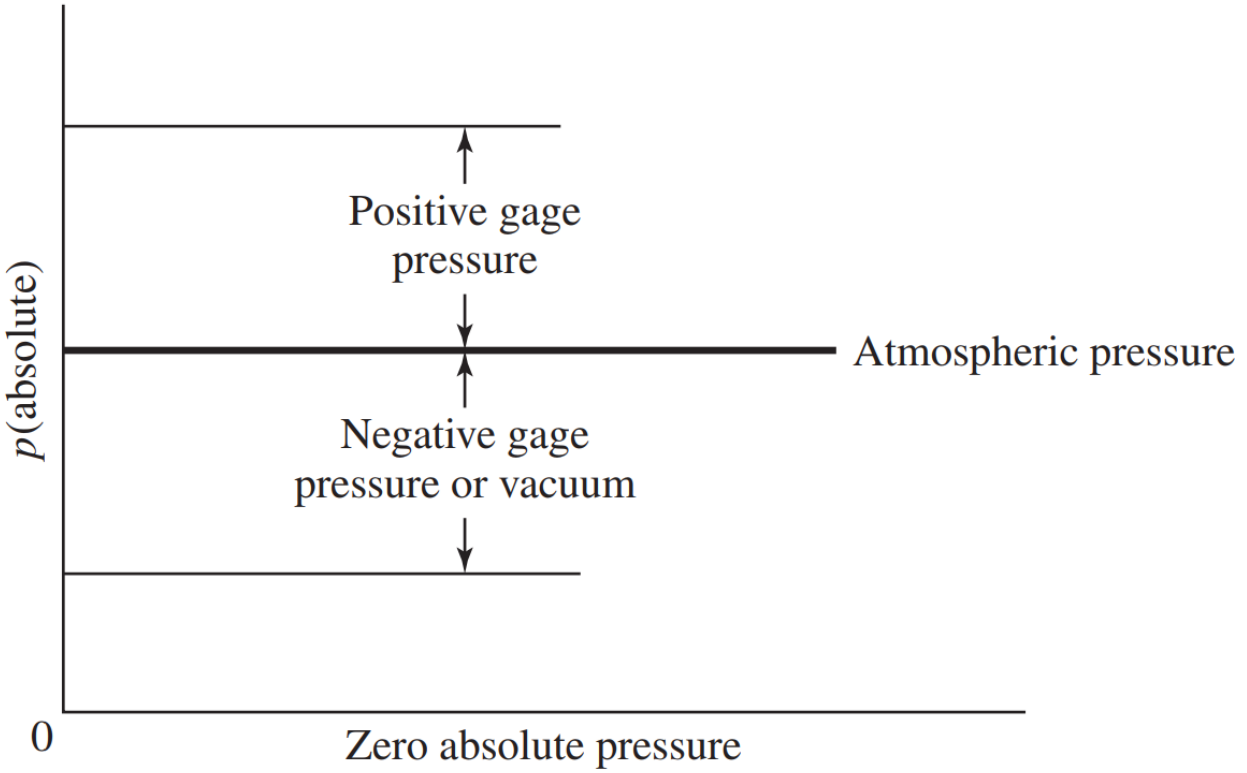
Absolute Pressure: The absolute force per unit area exerted by a fluid; it cannot be negative.

Gage Pressure: The difference between absolute pressure and local atmospheric pressure.

Vacuum: The amount by which local atmospheric pressure exceeds absolute pressure.

1 atm = 14.696 psia = 1.01325×10^5 Pa

= 760 mmHg = 1 bar.



Pressure Levels

Mechanical devices offer the simplest means for fluid pressure measurement under steady-state and laboratory conditions.

Pressure Measurement

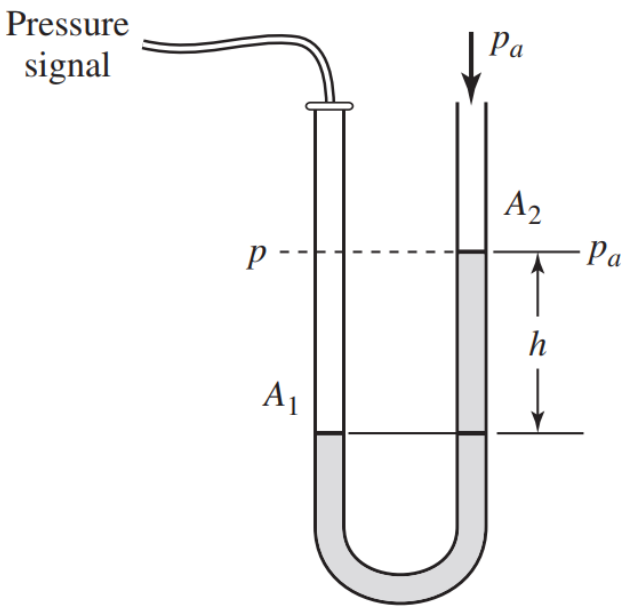
U-Tube Manometer Principles: Measures pressure difference $(p - p_a)$ as a function of the differential height h. Governing pressure balance equation:

$$p - p_a = gh(\rho_m - \rho_f)$$

where ρ_m is the manometer fluid density and ρ_f is the transmitting fluid density.

Sensitivity:

$$\text{Sensitivity} = \frac{h}{\Delta p} = \frac{1}{g(\rho_m - \rho_f)}$$



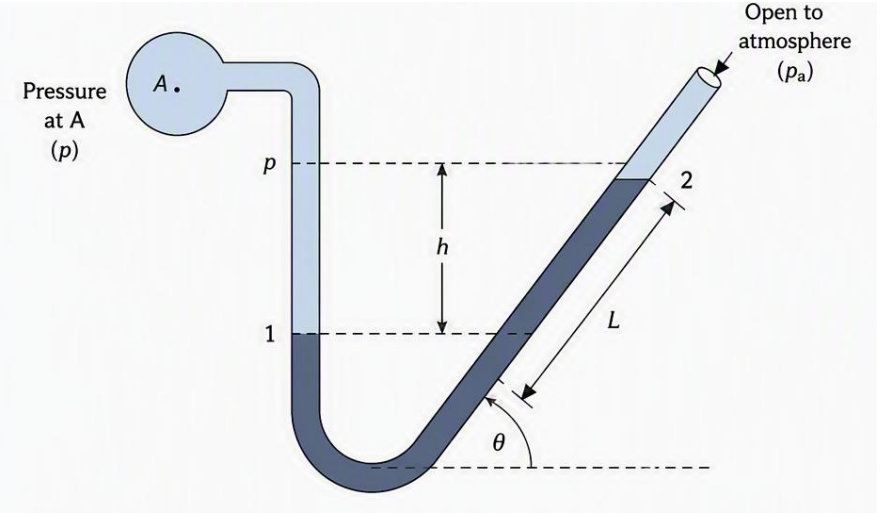
U-Tube Manometer

Inclined Manometer:

Lengthens the effective scale along an angle θ to significantly improv readability and sensitivity:

$$p - p_a = g(\rho_m - \rho_f)L \sin\theta$$

$$\text{Sensitivity} = \frac{1}{g \cdot \sin\theta \cdot (\rho_m - \rho_f)}$$



Inclined U-Tube Manometer

Pressure Measurement

Well-Type Manometer:

Features a large reservoir (A_1) and small column (A_2). Accounts for well level drop (y), yielding:

$$A_1 >> A_2$$

When pressure p is applied to the reservoir side:

Liquid level in the well falls slightly.

Liquid in the narrow tube rises by height h.

Let the fall in the well be y.

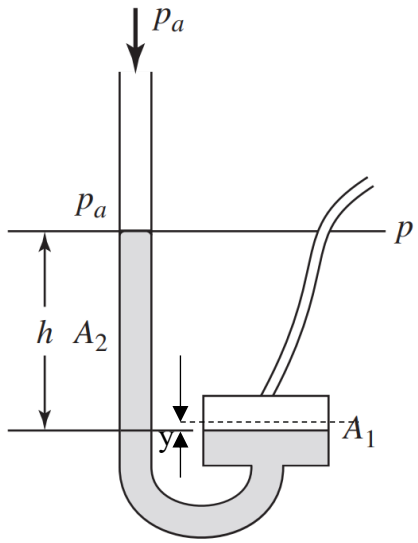
The rise in the tube is h.

By conservation of volume,

$$A_1 y = A_2 h$$

The actual vertical height difference between liquid levels is

$$H = h + y = h + \frac{A_2}{A_1} h$$



Well-Type Manometer

$$H = h\left(1 + \frac{A_2}{A_1}\right)$$

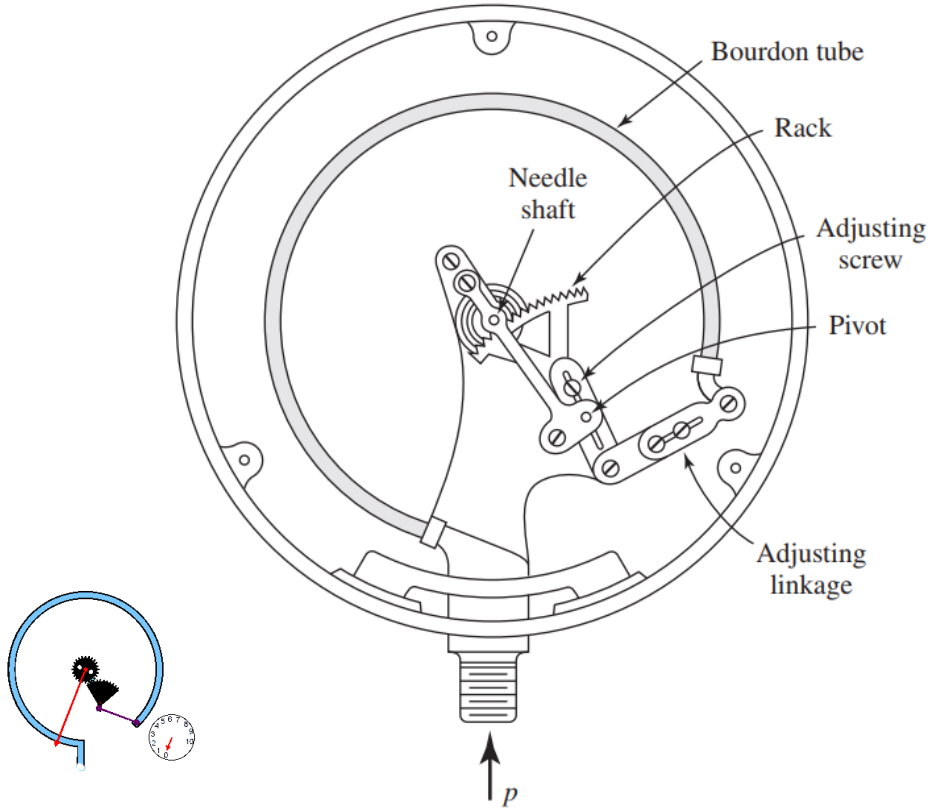
$$p - p_a = g(\rho_m - \rho_f)h\left(1 + \frac{A_2}{A_1}\right)$$

(commercial scales are pre-graduated to incorporate the area ratio factor).

Pressure Measurement

Bourdon Tube Pressure Gage:

- Primary Use: Widely utilized for consistent, inexpensive measurements of static pressure.
- Availability: Commercially available in sizes ranging from 1 to 16 inches in diameter with various accuracies.
- The Heise Gage: An extremely accurate Bourdon-tube gage with an accuracy of 0.1 percent of full-scale reading, frequently used as a reference pressure gage in laboratories.



Schematic of a bourdon-tube pressure gage

Construction & Operation

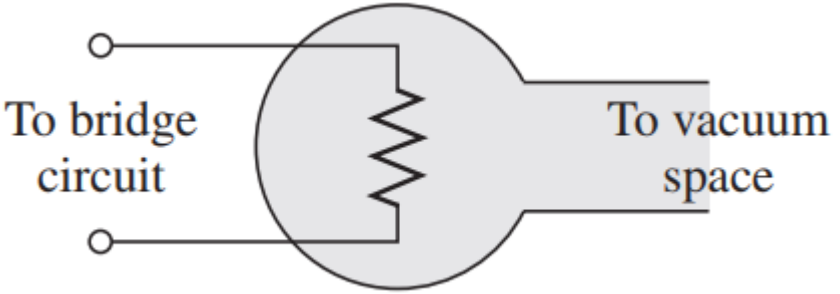
- Tube Design: Typically consists of an elliptical cross-sectional tube formed into a C-shape configuration.
- Working Principle: Applying internal pressure causes elastic deformation that is ideally proportional to the pressure.
- Mechanical Linkage: The free end connects to a spring-loaded linkage and rack mechanism that amplifies the displacement into an angular pointer rotation.
- Adjustability: Mechanisms can be tuned for optimal linearity, minimum hysteresis, and wear compensation.

Pressure Measurement

Pirani Thermal-Conductivity Gage:

Working Principle:

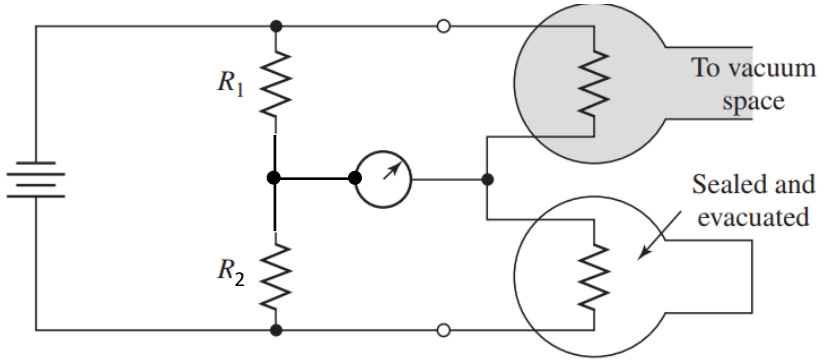
- Measures low pressures based on the principle that a gas's effective thermal conductivity decreases with pressure.
- An electrically heated filament loses heat to the surrounding gas; lower pressure reduces thermal conductivity, resulting in a higher filament temperature.
- Changes in filament temperature are measured by tracking the variation in its electrical resistance (using tungsten or platinum) typically via a bridge circuit.



Schematic of Pirani Gage

Ambient Compensation:

- Because heat loss also depends on ambient temperature, two gages (one evacuated reference, one active) are connected in series to compensate for environmental variations.



Pirani-gage arrangement to compensate for change in ambient temperature

Operating Range & Limitations:

- Typically effective between 0.1 to 100 Pa ($\approx 0.001\ Torr$ to 1 $Torr$).
- Requires empirical calibration.
- At very low pressures, a Pirani gauge hits a physical performance wall and experiences a sluggish response time due to specific thermodynamic constraints.

Pressure Measurement (Numerical)

A vessel contains water and its pressure is to be measured using both a vertical U-tube manometer and an inclined manometer.

- Density of manometer fluid (Mercury): $\rho_m = 13600\ kg/m^3$
- Density of transmitting fluid (Water): $\rho_f = 1000\ kg/m^3$
- Acceleration due to gravity: $g = 9.81\ m/s^2$
- Differential column height in U-tube manometer: $h = 0.15\ m$
- Inclined manometer angle: $\theta = 20°$

Determine:

- Pressure difference $(p - p_a)$using the U-tube manometer.
- Sensitivity of the U-tube manometer.
- Length of movement, L along the inclined tube for the same pressure difference.
- Sensitivity of the inclined manometer.
- Improvement in sensitivity achieved by using the inclined manometer.

Pressure Difference Using U-Tube Manometer:

$$\mathrm{p} - \mathrm{p_a} = \mathrm{gh}(\rho_\mathrm{m} - \rho_\mathrm{f})$$

Substituting values:

$$p - p_a = 9.81(0.15)(13600 - 1000)$$
$$= 9.81(0.15)(12600)$$
$$= 18540.9\ Pa$$
$$\boxed{p - p_a = 18.54\ kPa}$$

Sensitivity of U-Tube Manometer:

$$\mathrm{S_U} = \frac{\mathrm{h}}{\Delta \mathrm{p}} = \frac{1}{\mathrm{g}(\rho_\mathrm{m} - \rho_\mathrm{f})}$$
$$S_U = \frac{1}{9.81(12600)}$$
$$= 8.09 \times 10^{-6}\ m/Pa$$
$$\boxed{S_U = 8.09 \times 10^{-6}\ m/Pa}$$

or

$$\boxed{0.00809\ mm/Pa}$$

Pressure Measurement (Numerical)

A pressure vessel contains water and its pressure is to be measured using both a vertical U-tube manometer and an inclined manometer.

- Density of manometer fluid (Mercury): $\rho_m = 13600\ kg/m^3$
- Density of transmitting fluid (Water): $\rho_f = 1000\ kg/m^3$
- Acceleration due to gravity: $g = 9.81\ m/s^2$
- Differential column height in U-tube manometer: $h = 0.15\ m$
- Inclined manometer angle: $\theta = 20°$

Determine:

- Pressure difference $(p - p_a)$using the U-tube manometer.
- Sensitivity of the U-tube manometer.
- Length of movement Lalong the inclined tube for the same pressure difference.
- Sensitivity of the inclined manometer.
- Improvement in sensitivity achieved by using the inclined manometer.

Determine Length Along Inclined Tube:

For the inclined manometer:

$$p - p_a = g(\rho_m - \rho_f)L \sin\theta$$

Substituting:

$$18540.9 = 9.81(12600)L \sin 20°$$

Since

$$\sin 20° = 0.342$$

$$18540.9 = 42272.5L$$

$$L = 0.4385\ m$$

$$\boxed{L = 0.439\ m}$$

or

$$\boxed{438.5\ mm}$$

Sensitivity of Inclined Manometer:

$$S_I = \frac{1}{g \sin\theta\ (\rho_m - \rho_f)}$$

$$S_I = \frac{1}{9.81(0.342)(12600)}$$

$$= 2.37 \times 10^{-5}\ m/Pa$$

$$\boxed{S_I = 2.37 \times 10^{-5}\ m/Pa}$$

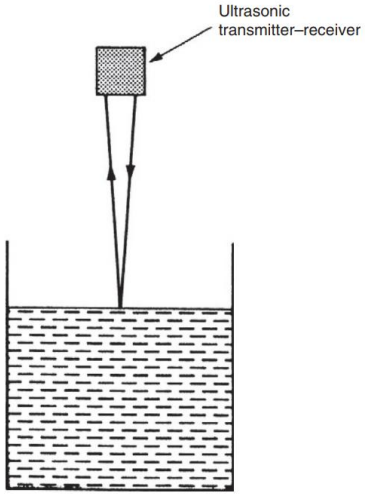
or

$$\boxed{0.0237\ mm/Pa}$$

Level Measurement

Ultrasonic Level Gauges:

- Noncontact technique that uses the time-of-flight of reflected sound waves to measure surface levels of viscous liquids or solids.
- Capable of measuring internal boundaries, such as the interface between two immiscible liquids or sludge/precipitate levels at the bottom of a tank.
- The working principle of the ultrasonic level gauge relies on measuring the time-of-flight of reflected sound waves:
- Signal Emission and Reflection: A transmitter-receiver unit mounted at the top of a tank emits high-frequency ultrasonic sound waves downward toward the target substance. When the sound waves strike the surface of the liquid or solid, they bounce back up as echoes to the receiver.
- Time-of-Flight Measurement: The sensor measures the exact time it takes for the sound pulse to make the round trip (down to the surface and back). By knowing the speed of sound in the medium, the system calculates the distance to the surface, thereby determining the level of the substance.
- Internal Boundary Detection: Beyond surface levels, it is also capable of measuring internal boundaries, such as the interface between two immiscible (non-mixing) liquids or the level of sludge/precipitates at the bottom of a tank.
- Temperature Compensation Requirement: Because the speed of sound through a gas or air medium changes with temperature, the system requires temperature compensation to maintain accuracy. With proper compensation, it can achieve an uncertainty of about ±1%.



Ultrasonic Level Gage

$$d = \frac{v \cdot t}{2}$$

Where:
d = the distance from the transducer to the surface of the substance.
v = speed of sound in the medium.
t = total round-trip time-of-flight of the sound wave (from emission to reception).

Level Measurement

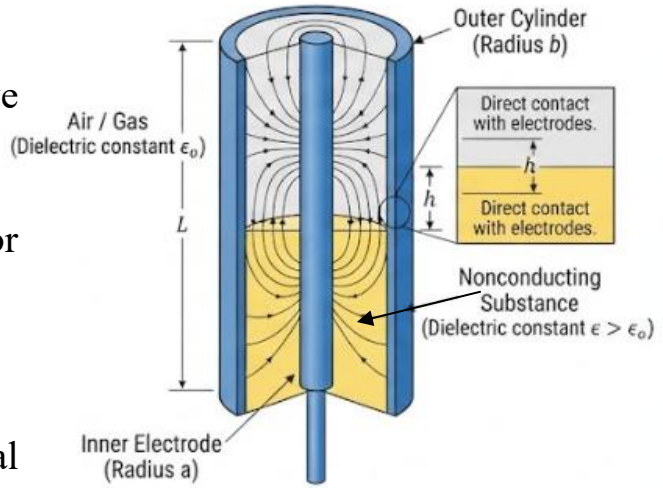
Capacitive Devices:

- Used for measuring liquid and solid (powder/granular) levels, including in extreme environments (corrosive liquids, liquid metals, high pressures).
- The sensor typically consists of two concentric electrodes—an inner rod of radius ‘a’ and an outer cylinder or tank wall of radius ‘b’.
- As the material level rises to a height ‘h’, it displaces the air or gas between the cylinders.
- Because the substance has a different electrical permittivity (dielectric constant) than air, the total electrical capacitance (C) of the system changes proportionally.
- Configurations depend on conductivity: Non-insulated concentric cylinders for nonconducting substances, and insulated concentric cylinders for conducting substances.
- Limitation: Prone to inaccuracy if moisture or contaminants alter the substance's dielectric constant.
- For concentric cylinder plates of radius ‘a’ and ‘b’ (b > a), and total height ‘L’, the depth of the substance ‘h’ is related to the measured capacitance ‘C’ by:

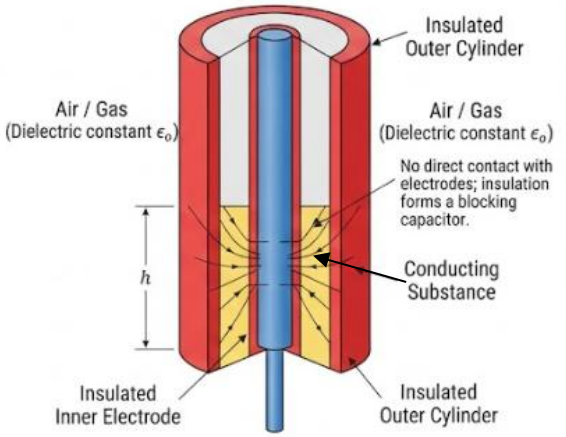
$$h = \frac{C\ln(b/a) - 2\pi\varepsilon_\circ L}{2\pi\varepsilon_\circ(\varepsilon - 1)}$$

where ε = relative permittivity of the measured non-conducting substance

$\varepsilon_\circ$ = permittivity of free space.



Capacitive Level Sensor for Non-Conducting Substance



Capacitive Level Sensor for Conducting Substance

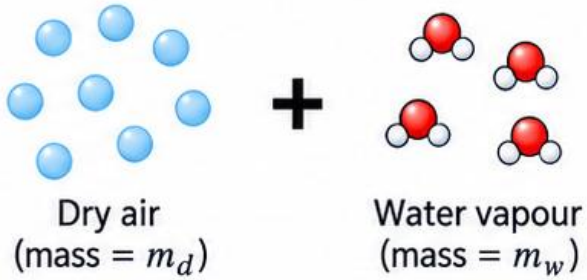
Concept of Humidity

Humidity is the amount of water vapour present in the air.

Ways to Express Humidity

Mixing ratio (humidity ratio), *r*

Mass of water vapour per unit mass of dry air.

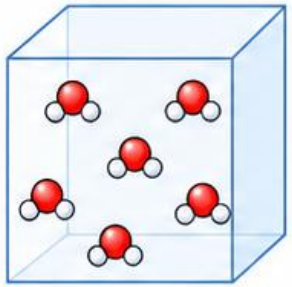


Dry air (mass = m_d) + Water vapour (mass = m_w)

$$r = \frac{\text{mass of water vapour}}{\text{mass of dry air}}$$

Absolute humidity

Mass of water vapour per unit volume of moist air.



$$d_w = \frac{m}{v}$$

m = mass of water vapour
v = volume of moist air

Relative humidity (RH)

Ratio of the actual water-vapour pressure to the saturation vapour pressure, expressed as a percentage.

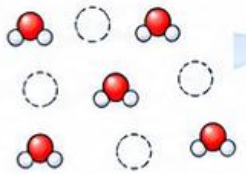
$$RH = \frac{P_w}{P_s} \times 100$$

P_w = actual water-vapour pressure
P_s = saturation vapour pressure at the same temperature

Example: If $P_s = 4$ kPa and $P_w = 2$ kPa, then

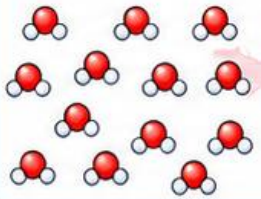
$$RH = \left(\frac{2}{4}\right) \times 100 = \mathbf{50\%}$$

RH = 50%



Room for more vapour (before saturation).

RH = 100% (saturated air),



Condensation/ dew can start.

At constant moisture content: Temperature ↑ → saturation pressure increases → RH decreases.

Relative humidity tells how close the air is to saturation.

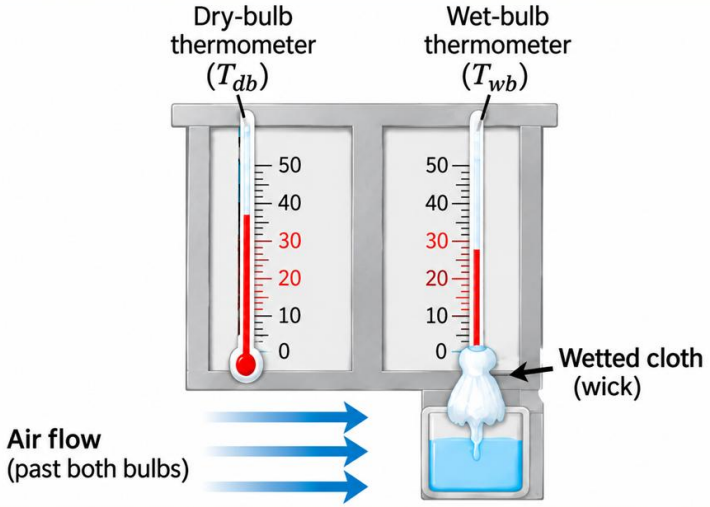
Humidity Measurement using Dry- and Wet-Bulb Temperatures

Principle

- Relative humidity can be determined using the **temperature of dry-bulb thermometer and a wet-bulb thermometer**.
- The wet-bulb thermometer bulb is covered with a wetted cloth. Water evaporates from the cloth, absorbing heat and lowering the thermometer reading.
- The drier the air, the higher the rate of evaporation and the greater the difference $(T_{db} - T_{wb})$.

$$T_{db} - T_{wb} \propto \frac{1}{RH}$$

Dry- and Wet-Bulb Thermometers

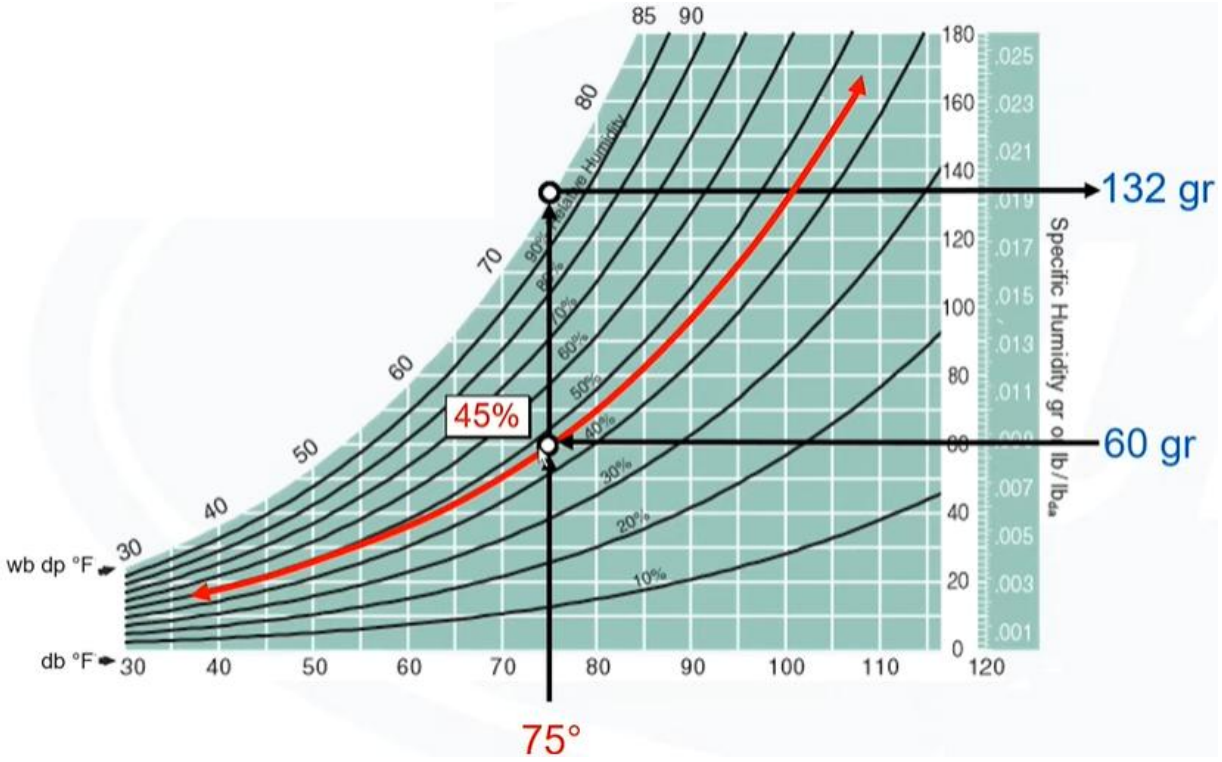


Important points

- T_{db} is the actual air temperature.
- $T_{wb} \leq T_{db}$.
- At 100% RH, $T_{db} = T_{wb}$.
- Larger $(T_{db} - T_{wb})$ indicates drier air.

Using the psychrometric chart

For a given atmospheric pressure, **RH can be obtained from a psychrometric chart or empirical correlations using T_{db} and T_{wb}.**



Finding Relative Humidity

- Saturation vapour pressure:

$$P_s(T) = 10^{\left(10.23 - \frac{1750.2}{0.999 * T - 37.88}\right)}$$

Where T is temperature in K and P_s is pressure in Pa.

- The actual vapor pressure can be obtained from the correlation:

$$P_w = P_{s\,@\,Twb} - A \times P \times (T_{db} - T_{wb})$$

- For natural air circulation: $A = \frac{0.8}{1000}$

P is in Pa, A is in K^{-1} (Where P is atmospheric Pressure)

- For force- air circulation : $A = \frac{0.667}{1000}$

- Relative Humidity= $\frac{\text{Actual vapor pressure}}{\text{Saturation vapor pressure at dry bulb temperature}}$

Ref: Guo et.al (2022), *Applied Gaseous Fluid Drilling Engineering* (pp. 15–28), Elsevier. *Ref: Fraden, Ch. 13 (Humidity and Moisture Sensors).*

Problem 1

A naturally ventilated psychrometer gives a **dry-bulb temperature of** 32°C and a **wet-bulb temperature of** 24°C. Determine Relative humidity.

Solution

Given:

$$T_{db} = 32°C = 305.15\,K$$
$$T_{wb} = 24°C = 297.15\,K$$

For natural air circulation,

$$A = \frac{0.8}{1000} = 0.0008\ K^{-1}$$

Assume standard atmospheric pressure,

$$P = 101325\ Pa$$

1. Saturation vapour pressure at wet-bulb temperature

Using

$$P_s(T) = 10^{\left(10.23 - \frac{1750.2}{0.999T - 37.88}\right)}$$

At $T_{wb} = 297.15\,K$,

$$P_{s@T_{wb}} = 10^{\left(10.23 - \frac{1750.2}{0.999(297.15) - 37.88}\right)}$$
$$\boxed{P_{s@T_{wb}} \approx 2963.2\ Pa}$$

2. Actual vapour pressure

$$P_w = P_{s@T_{wb}} - AP(T_{db} - T_{wb})$$
$$P_w = 2963.2 - (0.0008)(101325)(305.15 - 297.15)$$

Temperature difference:

$$305.15 - 297.15 = 8\,K$$

Therefore,

$$P_w = 2963.2 - (0.0008)(101325)(8)$$
$$P_w = 2963.2 - 648.48$$
$$\boxed{P_w \approx 2314.7\ Pa}$$

3. Saturation vapour pressure at dry-bulb temperature

At $T_{db} = 305.15\,K$,

$$P_{s@T_{db}} = 10^{\left(10.23 - \frac{1750.2}{0.999(305.15) - 37.88}\right)}$$
$$\boxed{P_{s@T_{db}} \approx 4721.5\ Pa}$$

4. Relative humidity

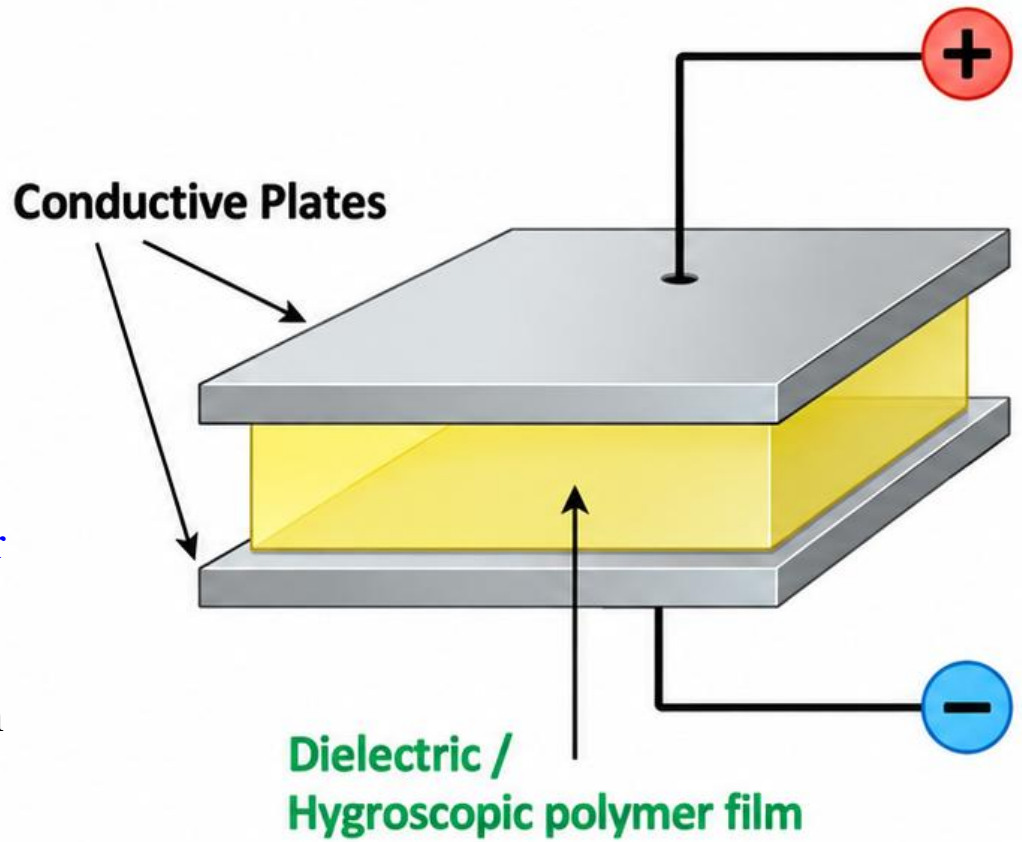
$$RH = \frac{\text{Actual vapour pressure}}{\text{Saturation vapour pressure at dry−bulb temperature}}$$
$$RH = \frac{2314.7}{4721.5}$$
$$RH = 0.4903$$

Therefore,

$$\boxed{\text{RH} \approx 49.0\%}$$

Capacitive Hygrometers

- A capacitive humidity sensing device is designed with a polymer film sandwiched between metal conductive electrodes.
- The capacitive relative humidity (RH) sensor is formed by merging two conductive plates that sandwich an insulator referred to as a dielectric.
- The capacitance depends on the area of both plates, the separation between them, and the dielectric constant of the material.
- The purpose of the dielectric (insulator) is to absorb any water vapour diffusing through the RH sensor.
- Hence, the space between the capacitor plates is usually filled with an appropriate dielectric material whose dielectric constant varies when it is subjected to change in humidity.
- The common method of constructing a capacitive RH sensor is to use a hygroscopic polymer film as dielectric and deposit two layers of electrodes on either side.

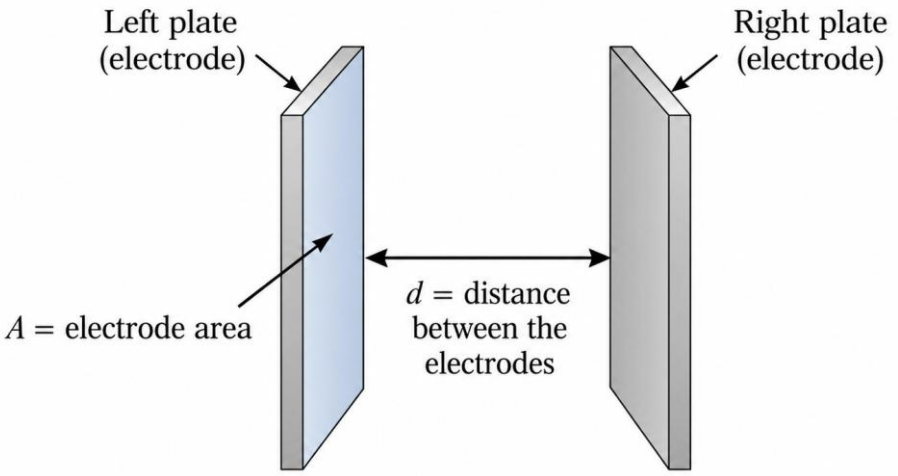


Capacitive Hygrometers

1. The **absolute permittivity** of moist air is related to its relative permittivity

$$\varepsilon = \kappa\varepsilon_0$$

where ε_0 is the permittivity of free space (unit: F/m , farad per meter).
The value of the permittivity of free space is: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m
κ is the relative permittivity, also called the dielectric constant of moist air.



2. For an air-filled parallel-plate capacitor, the capacitance is

$$C = \frac{\varepsilon A}{d} = \frac{\kappa\varepsilon_0 A}{d}$$

where A is the electrode area and d is the distance between the electrodes.

3. **Humidity changes the electrical permittivity of the air.** The relative permittivity or dielectric constant, κ, of moist air is given by :

$$\kappa = 1 + \frac{211}{T}\left(\frac{P}{133.3} + \frac{4800 * P_s}{T * 133.3}H\right)10^{-6}$$

where:
T= absolute temperature, K
P= pressure of moist air, Pa
P_s= saturation vapour pressure of water at temperature T, Pa
H= relative humidity.

Problem 2

A capacitive relative humidity (RH) sensor is constructed using a polymer film sandwiched as a dielectric medium between two metal conductive electrodes each of area 100 cm^2 and separated by a distance of 1 mm. Water vapor in the air diffuses into the polymer film, altering its dielectric constant (κ) according to the relationship $\kappa = 1 + \frac{211}{T}\left(\frac{P}{133.3} + \frac{4800*P_s}{T*133.3}H\right)10^{-6}$, where T is absolute temperature, P is total pressure, P_s is saturation vapor pressure, and H is relative humidity. At a room temperature of 25°C and atmospheric pressure of $101.3 * 10^3$ Pa and $P_s = 3{,}167.7$ Pa, the sensor capacitance is measured to be 88.602 pF. calculate the relative humidity (H) sensed by the device.

Solution

$$\boxed{\kappa = 1 + \frac{211}{T}\left(\frac{P}{133.3} + \frac{4800P_s}{T(133.3)}H\right)10^{-6}}$$

and the capacitor relation

$$\boxed{C = \frac{\kappa\varepsilon_0 A}{d}}$$

we can solve for H.

1. Given data

$$A = 100\ \text{cm}^2 = 0.01\ \text{m}^2$$
$$d = 1\ \text{mm} = 0.001\ \text{m}$$
$$C = 88.602\ \text{pF} = 88.602 \times 10^{-12}\ \text{F}$$
$$\varepsilon_0 = 8.854 \times 10^{-12}\ \text{F/m}$$
$$T = 25 + 273.15 = 298.15\ \text{K}$$
$$P = 101.3 \times 10^3 = 101300\ \text{Pa}$$
$$P_s = 3167.7\ \text{Pa}$$

2. Find dielectric constant κ

From

$$C = \frac{\kappa\varepsilon_0 A}{d}, \qquad \kappa = \frac{Cd}{\varepsilon_0 A}$$

Substituting,

$$\kappa = \frac{(88.602\times10^{-12})(0.001)}{(8.854\times10^{-12})(0.01)}$$

$$\boxed{\kappa = 1.00070025}$$

3. Substitute into the humidity equation

$$1.00070025 = 1 + \frac{211}{298.15}\left[\frac{101300}{133.3} + \frac{4800(3167.7)}{298.15(133.3)}H\right]10^{-6}$$

First,

$$\frac{101300}{133.3} = 759.940$$

and

$$\frac{4800(3167.7)}{298.15(133.3)} = 382.578$$

Therefore,

$$0.000700248 = \frac{211}{298.15}(759.940 + 382.578H)10^{-6}$$

Rearranging,

$$759.940 + 382.578H = 989.474$$
$$382.578H = 989.474 - 759.940$$
$$382.578H = 229.534$$
$$H = 0.599967$$

Therefore,

$$\boxed{H = 0.6}$$

or, in percentage ,

$$\boxed{\text{Relative Humidity} = 60\%}$$